Q1. The Chrono-Locked Conveyor Belt

In an ancient mechanical city, a circular conveyor belt transports crystalline energy cores in a strictly non-decreasing sequence of stability levels. The belt was originally calibrated in sorted order to prevent resonance collapse.

However, during a maintenance cycle, the system was restarted from an arbitrary position on the belt. As a result, the visible sequence now appears rotated, though it must still preserve the integrity of the original sorted structure.

The engineers suspect that:

- The belt was originally arranged in **non-decreasing order**.
- It may have been rotated **any number of positions (including zero)**.
- Duplicate stability levels are allowed.
- If the belt does not conform to such a rotated sorted configuration, the stabilization chamber will fail.

Your task is to determine whether the currently observed sequence could represent a valid rotation of a non-decreasing sorted array.

Return **true** if valid. Otherwise, return **false**.

Test Cases

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Input: [5,6,7,1,2,3,4]
Output: true

Input: [2,2,2,3,1,2]
Output: true

Input: [1,3,2,4]
Output: false

Input: [10,10,10]
Output: true

Input: [4,5,1,2,6]
Output: false

Q2. The Genetic Archive Compression Protocol

In a bioinformatics vault, DNA fragments are stored in a sequence sorted by genetic marker intensity (non-decreasing). Due to replication errors, several identical fragments exist consecutively.

The archive has limited storage capacity. The system requires:

- Each distinct genetic fragment must appear only once.
- The relative order must remain unchanged.

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- All modifications must be performed **in-place**.
- Only the first **k** positions must contain the unique fragments.
- The rest of the array beyond index **k-1** is irrelevant.

Return the number **k**, representing the count of unique fragments remaining.

Test Cases

Input: [1,1,1,2,2,3]
Output: 3

Input: [-3,-3,-2,-1,-1,0,0,0,1]
Output: 5

Input: [5,5,5,5,5]
Output: 1

Input: [0,1,2,3,4,5]
Output: 6

Input: [-10,-10,-5,-5,-5,-2,-1,-1]
Output: 4

Q3. The Vanished Identification Code

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A security vault issues encrypted identification codes ranging from **0 to n** inclusive. Exactly **n** codes are active in the system, but due to a cyber breach, one code has disappeared.

You are given an array containing **n distinct numbers** within the range **[0, n]**.

Your mission:
Identify the missing identification code.

Return the missing number.

Test Cases

Input: [0,1,3,4,5]
Output: 2

Input: [1]
Output: 0

Input: [0]
Output: 1

Input: [4,2,1,0]
Output: 3

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Input: [8,6,4,2,3,5,7,0,1]
Output: 9

Q4. The Zero-Gravity Cargo Rebalancing

A spacecraft cargo deck records mass distribution in an integer array. Due to anti-gravity calibration errors, all cargo units with mass zero must be relocated to the end of the deck.

Constraints:

- Relative order of non-zero cargo must remain unchanged.
- Operation must be performed **in-place**.
- No additional cargo deck may be allocated.

Rearrange the array so that all zeros appear at the end.

Test Cases

Input: [0,0,1,0,3,12]
Output: [1,3,12,0,0,0]

Input: [4,2,4,0,0,3,0,5,1,0]
Output: [4,2,4,3,5,1,0,0,0,0]

Input: [0,0,0]
Output: [0,0,0]

Input: [1,2,3]
Output: [1,2,3]

Input: [0,1,0,0,2,0,3,4]
Output: [1,2,3,4,0,0,0,0]

Q5. The Tri-Color Signal Sorting Mechanism

Inside a quantum communication facility, each signal packet is labeled with a priority code:

- 0 → Low Priority
- 1 → Medium Priority
- 2 → High Priority

Due to electromagnetic interference, the packets are scrambled. The control system demands:

- Sort the packets in ascending order of priority.
- Must be done in **O(n)** time.
- Must use **O(1)** extra space.

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- No secondary storage arrays allowed.

Restore the signal stream to proper priority order.

Test Cases

Input: [2,0,1,2,1,0,0,2]
Output: [0,0,0,1,1,2,2,2]

Input: [1,1,1,1]
Output: [1,1,1,1]

Input: [0,2,2,1,0,1,2,0,1]
Output: [0,0,0,1,1,1,2,2,2]

Input: [2,2,2,2,0]
Output: [0,2,2,2,2]

Input: [0]
Output: [0]

Q6. The Signal Stability Analyzer

In a deep-space transmission relay, binary signals are streamed continuously.

- 1 represents a stable signal pulse.
- 0 represents a disruption caused by cosmic interference.

The stability of a transmission session is measured by the **longest uninterrupted chain of stable pulses**.

Your task is to determine the maximum number of consecutive 1s in the signal log.

Test Cases

Input: [1,1,0,1,1,1,0,1]
Output: 3

Input: [0,0,0,0]
Output: 0

Input: [1,1,1,1,1]
Output: 5

Input: [1,0,1,1,0,1,1,1,1,0,1]
Output: 4

Input: [0,1,1,0,1,1,1,0,1,1]
Output: 3

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Q7. The Mountain Watchtower Protocol

A kingdom has watchtowers built along a mountain ridge. Each tower has a defensive strength value.

A tower is considered a **Guardian Tower** if no tower to its right has greater strength.
The rightmost tower is always a Guardian Tower.

Your task: Identify all Guardian Towers in the given sequence.

Test Cases

Input: [16,17,4,3,5,2]
Output: [17,5,2]

Input: [10,4,2,4,1]
Output: [10,4,4,1]

Input: [5,10,20,40]
Output: [40]

Input: [30,20,20,10]
Output: [30,20,20,10]

Input: [7,7,7,7]
Output: [7,7,7,7]

Q8. The Rotational Cipher Machine

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An encryption engine stores a sequence of integers. To activate a new cipher pattern, the entire sequence must be rotated to the right by k positions.

Rotation Rule:
Each element shifts right by one index, and elements that fall off reappear at the beginning.

You must rotate the array in-place.

Test Cases

Input: nums = [1,2,3,4,5,6,7], k = 3
Output: [5,6,7,1,2,3,4]

Input: nums = [-1,-100,3,99], k = 2
Output: [3,99,-1,-100]

Input: nums = [1,2], k = 5
Output: [2,1]

Input: nums = [10,20,30], k = 0
Output: [10,20,30]

Input: nums = [4,5,6,7,8], k = 7
Output: [7,8,4,5,6]

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Q9. The Cross-Domain Intelligence Merge

Two intelligence agencies maintain distinct encrypted datasets containing unique numerical codes.

You must determine how many distinct codes are common between both datasets.

Return only the count of unique intersecting elements.

Test Cases

Input: a = [89,24,75,11,23], b = [89,2,4]
Output: 1

Input: a = [1,2,3,4,5,6], b = [3,4,5,6,7]
Output: 4

Input: a = [10,20,30], b = [40,50,60]
Output: 0

Input: a = [5,10,15,20], b = [5,10,15,20]
Output: 4

Input: a = [1,3,5,7,9], b = [2,4,6,8,10]
Output: 0

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Q10. The Equilibrium Sentinel

A sequence of power nodes is arranged randomly.

You must find the **first node** such that:

- Every node to its left has strictly smaller value.
- Every node to its right has strictly greater value.

If no such node exists, return -1.

Test Cases

Input: [4,2,5,7]
Output: 5

Input: [11,9,12]
Output: -1

Input: [1,2,3,4,5]
Output: 2

Input: [5,1,4,4]
Output: -1

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Input: [2,3,8,9,10,5]
Output: -1

Q11. The Frequency Reconstruction Grid

A corrupted database stores integers between 1 and p.
You are given:

- n (size for counting range 1 to n)
- arr[]
- p (upper possible bound of values)

You must modify arr[] in-place such that:
arr[i] becomes the frequency of integer (i+1).

Ignore values greater than n.

Test Cases

Input: n=5, arr=[2,3,2,3,5], p=5
Output: [0,2,2,0,1]

Input: n=4, arr=[3,3,3,3], p=3
Output: [0,0,4,0]

Input: n=6, arr=[1,2,3,4,5,6], p=6
Output: [1,1,1,1,1,1]

Input: n=3, arr=[4,5,6], p=6
Output: [0,0,0]

Input: n=7, arr=[1,1,2,8,2,3,10], p=10
Output: [2,2,1,0,0,0,0]

Q12. The Resource Amplification Strategy

You are given resource values in an array.
In one operation, you may increment any element by 1.

You may perform at most k operations.

Determine the maximum possible frequency of any single value after operations.

Test Cases

Input: nums=[1,2,4], k=5
Output: 3

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Input: nums=[1,4,8,13], k=5
Output: 2

Input: nums=[3,9,6], k=2
Output: 1

Input: nums=[1,1,1,2,2,4], k=3
Output: 5

Input: nums=[5,5,5,5], k=10
Output: 4

Q13. The Rat Colony Survival Simulation

A colony of r rats requires unit amount of food per rat.

An array represents food available in consecutive houses.

Determine the **minimum number of houses** required to feed all rats.

Return:

- -1 if array is null
- 0 if total food is insufficient

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Test Cases

Input: r=7, unit=2, arr=[2,8,3,5,7,4,1,2]
Output: 4

Input: r=5, unit=3, arr=[1,2,3,4]
Output: 0

Input: r=3, unit=5, arr=[10,5,5]
Output: 2

Input: r=1, unit=10, arr=[10]
Output: 1

Input: r=4, unit=2, arr=null
Output: -1

Q14. The Minimal Comparison Duel

In a competitive tournament, participants' scores are recorded in an array.

You must determine:

- Minimum score
- Maximum score

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Using the **minimum number of comparisons** possible.

Test Cases

Input: [3,5,4,1,9]
Output: Min=1, Max=9

Input: [22,14,8,17,35,3]
Output: Min=3, Max=35

Input: [100]
Output: Min=100, Max=100

Input: [-5,-1,-9,-3]
Output: Min=-9, Max=-1

Input: [7,7,7,7]
Output: Min=7, Max=7

Q15. The Hourglass Energy Matrix

A 6×6 energy grid contains both positive and negative charge values.

An hourglass is defined as:

a b c
  d
e f g

There are exactly 16 possible hourglasses.

Compute the maximum hourglass sum.

Test Cases

Input:
1 1 1 0 0 0
0 1 0 0 0 0
1 1 1 0 0 0
0 0 2 4 4 0
0 0 0 2 0 0
0 0 1 2 4 0

Output: 19

Input:
-9 -9 -9 1 1 1
0 -9 0 4 3 2
-9 -9 -9 1 2 3

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0 0 8 6 6 0
0 0 0 -2 0 0
0 0 1 2 4 0

Output: 28

Input:
0 0 0 0 0 0
0 0 0 0 0 0
0 0 0 0 0 0
0 0 0 0 0 0
0 0 0 0 0 0
0 0 0 0 0 0

Output: 0

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